AgriChill Advisor

A software-led cold-room advisory for farmer groups: project report for IEEE Smart Village IdeaRun 2026

Status: DRAFT for team discussion, 13 October 2026. Items marked [VERIFY] are assumptions that must be confirmed in the field before you submit. The official deadline on the IdeaRun page is 31 October 2026; the competition's funding amount and judging criteria are not published there.

1. Summary in plain words

Farmers in the same area often harvest the same crop in the same week. When too much produce reaches the market at once, prices crash. Fruits and vegetables spoil quickly, so farmers cannot wait; many sell at a loss or throw the produce away. A cold room lets a farmer hold produce for a few days, but small cold rooms run by farmer groups need tools: who is booking space, how long each batch will stay good, when to sell, and how to use electricity (especially solar) efficiently.

AgriChill Advisor is a simple app, in the local language, for a farmer group and its cold room. It handles bookings, watches storage conditions through off-the-shelf sensors, estimates how many good days each batch has left, suggests when to hold or sell using past market prices, and shows the electricity cost per crate. It is advice for people, not an automatic decision-maker.

2. A real-world scenario

The farmer below is a fictional composite, invented to explain the problem. The background facts about tomato price crashes are from public reports; the numbers in the story are illustrative, not data.

Murugan, a tomato farmer in Krishnagiri district, Tamil Nadu

Background that is documented: in 2019, farmers in Rayakottai (Krishnagiri district) and Valappadi (Salem district) dumped tomatoes into lakes after prices fell from about Rs 20 to Rs 2 per kg. A union leader blamed many farmers choosing tomato, a short-season crop. A report said rural cold storage was insufficient and that tomatoes have a low shelf life (sources in section 11).

A day WITHOUT the system

- Murugan harvests 3 tonnes. The same week, most farmers around him harvest too. The mandi is full and the trader offers a very low price.
- A farmer-group cold room exists 20 km away, but Murugan does not know if space is free, what it costs, or how long his tomatoes would last in it.
- The cold-room operator takes bookings on paper and phone calls. Some crates are accepted too late, and the operator cannot tell which batches will spoil first.
- Murugan sells below his cost, or throws part of the harvest away.

A day WITH the system (what we propose to build)

- Murugan sends a message in Tamil to the farmer group's number and books 40 crates for three days. He sees the price per crate.
- A temperature and humidity logger in the cold room feeds the app. Each batch gets an estimate of remaining good days, shown as a range, for example 'about 4 to 6 days'.

- The app shows past market prices for his crop and tells him, in plain words, how prices moved after similar glut weeks in earlier seasons, with a clear note that this is not a promise.
- The operator sees which batches to sell first, how full the room is, and the electricity cost per crate. The app suggests cooling at times when cheaper or solar power is available.
- Murugan decides when to sell. Over a season, the group compares spoilage and income with the previous season.

Honest limit: if market prices do not recover, holding produce does not help. The tool must show uncertainty, not false confidence.

3. Where and why the problem occurs

3.1 Evidence we found

Fact	Source	How reliable
Annual harvest and post-harvest losses of major produce were estimated at Rs 92,651 crore (CIPHET study, 2012-13 production at 2014 prices).	Government reply to Parliament	Official, but dated
Ministry figures cited for losses: fruits Rs 16,644 crore, vegetables Rs 14,842 crore.	Global Agriculture, citing a 2021 Lok Sabha reply	Official figures via a news site
Press reports give a different figure for fruit and vegetables (about 16% lost, Rs 40,811 crore).	IndiaSpend and FreshPlaza	Conflicts with the above. Use the official figures and state the conflict
Tomato prices fell from about Rs 20 to Rs 2 per kg and farmers dumped produce in Rayakottai (Krishnagiri) and Valappadi (Salem), 2019.	The News Minute, Krishi Jagran	News reports, 2019. Needs a current check
High up-front cost has held back solar cold storage; business-model pilots are recommended.	SELCO Foundation landscape study terms	NGO document
Pay-as-you-use cold storage through FPOs is emerging, with Andhra Pradesh and Maharashtra named as early adopters.	Mongabay India, Nov 2025	News report on an NCCD report
Cold storage capacity is unevenly distributed across states.	A BASE/Empa project note found in search; RBI publishes a state-wise table	Not read in full. Pull the RBI table for your state

3.2 Why it happens (causes)

- Same-time harvesting and short-season crops create gluts and price crashes.
- Fruits and vegetables are perishable, so farmers cannot wait for better prices.
- Cold storage is limited in rural areas and unevenly spread across states.
- Solar or efficient cold rooms cost a lot up front, so small groups hesitate.
- Small operators and farmers lack simple tools for booking, spoilage risk and sell timing. [VERIFY with 5 farmers and 1 operator: this is our hypothesis, not a proven gap.]

3.3 Which state or district?

The problem is national, but the evidence differs by place. Documented examples: tomato gluts in Krishnagiri and Salem (Tamil Nadu), and pay-per-use cold storage through FPOs in Andhra Pradesh and Maharashtra. Choose your pilot district by where you can get a partner (a farmer group or cold-room owner), not by headline statistics. We do not yet know your district; please send it so this section can be made specific.

4. Who the project is for, and who benefits

Group	Problem today	What could change	How we would measure it
Smallholder farmers, especially members of farmer producer organisations (FPOs)	Forced to sell at harvest-time prices; spoilage	Sell later, lose less	Spoilage %, income per quintal vs last season or vs a comparison group
FPO or cold-room operator	Empty or badly planned space; unclear costs	Better bookings, a clear price per crate, lower power cost	Room utilisation %, kWh per quintal, revenue per month
Self-help group members and small aggregators handling produce	Same price and storage problems	Shared access through the group's agent or phone	Number of users, repeat bookings
Local agriculture advisers (for example agricultural university and Krishi Vigyan Kendra staff) [VERIFY]	Advice is general and not tied to a batch	Local data they can use	Feedback from advisers
Consumers and local traders (indirect)	Price swings	Possibly steadier supply	Not measured by this project

Who may not benefit: farmers without phones or far from a cold room, and larger farmers who adopt faster. We should design for inclusion: voice and local-language messages, access through the group's agent, and a price that smaller farmers can afford.

5. What we build

Module	What it does	AI / method	Data needed
Booking and inventory	Book crates, track batches, show price per crate; Tamil or local language	None needed, plus an optional language assistant	Farmer group's records
Spoilage-risk estimate	Estimate remaining good days for each batch, as a range	Simple probabilistic model using crop shelf-life tables and logger readings, updated as data arrives	Logger data; shelf-life tables from agricultural research bodies [VERIFY source]; manual grading of batches
Hold-or-sell guidance	Show past price patterns and arrival trends for the crop	Time-series forecasting with honest error ranges	Daily market prices (see note below)
Energy view	Show kWh per crate and suggest cooling times	Rule-based plus a solar or tariff forecast	A commercial clip-on energy meter and tariff

Why simple models: with one season of small data, a deep neural network cannot be validated and cannot be explained to farmers. A transparent probabilistic model with uncertainty ranges is easier to defend.

Market-price data note: the daily mandi price dataset on data.gov.in is generated from the Agmarknet portal, but its page says no API exists and asks users to request one; a historical archive from Ashoka University (CEDA) exists but needs a key. Check access, terms and limits before relying on either.

6. How we prove it works

- Month 1 baseline: record last season's spoilage, selling prices and cold-room utilisation for the same group.
- Pilot for one harvest season. Compare with the baseline and, if possible, with nearby non-users.
- Metrics: spoilage %, income per quintal, room utilisation %, kWh per quintal, number of active users, and a short usability survey.
- Be honest in the report: one season and one site cannot prove cause and effect. Say what the result does and does not show.

7. Plan and budget structure (6 to 12 months)

Period	Work
Month 1	Interviews, partner agreement, baseline
Months 2-3	Build the minimum version; test with the operator
Months 4-6	Pilot through one harvest season; fix problems
Months 7-9	Improve; add a second site or crop if possible
Months 10-12	Results report; release the code and a how-to guide openly

Budget lines to price with real quotes: temperature and humidity loggers, energy meters, messaging or SMS costs, field travel, cloud hosting, training sessions for farmers, and printing. Past reported grants were about $2,000 to $2,982 (a college report); do not assume more unless the brochure says so.

8. Risks and ethics

- No partner: the project fails without a farmer group or cold-room owner. Get a signed letter before submitting.
- Season mismatch: your crop's harvest may not fit the timeline. Choose a crop with a season in your window, or a year-round chain such as milk.
- Farmer data: get consent, collect only what is needed, store it safely, and explain in Tamil how it is used.
- Advice liability: label output as guidance with uncertainty, never a guarantee of profit.
- Equipment: use ready-made loggers; do not claim hardware design skills you do not have. Consider adding an EEE or ECE teammate (teams of up to 5 are allowed).

9. What is verified and what is not

Verified from sources	Not verified: confirm before submitting
IdeaRun 2026 theme, team rules, 1000-word limit, 6-12 month duration, 31 October deadline	The ₹10 lakh figure (not on the page), the 2026 funding amount, the judging criteria
National loss estimates (official replies); 2019 tomato price crash in Krishnagiri and Salem	That small operators lack booking and spoilage tools in your district
Solar cold-storage cost barrier; pay-per-use FPO models in AP and Maharashtra	Whether your region has a suitable cold room and farmer group
Mandi price data exists, with API access restrictions	Shelf-life values for your crop, logger costs, and software-

	only eligibility

10. Next steps before the deadline

- Today: email idearun.isvsarc@gmail.com to confirm funding, deadline and whether software-led proposals are eligible. Ask your faculty mentor whether your college has an earlier internal deadline.
- Days 1-4: choose a district and crop; find one farmer group or cold-room owner; get a short letter of support.
- Days 4-7: interview 5 farmers and the operator; record a baseline.
- Days 8-12: write the 1000-word proposal from this report; get quotes for the budget.
- Days 13-16: faculty mentor review. Submit by about 28 October, not on the last day.

11. Sources

- IdeaRun 2026 page: https://isvsymposium.org/idea-run
- CIPHET losses estimate, Lok Sabha reply: https://eparlib.nic.in/bitstream/123456789/706362/1/56899.pdf
- Fruit and vegetable loss figures: https://www.global-agriculture.com/national-news/scheme-for-integrated-cold-chain-and-value-addition-infrastructure-in-operation/
- Conflicting press figure: https://indiaspend.com/indias-agriculture-produce-losses-rs-92000-cr-3-times-more-than-agriculture-budget-71003
- Tomato dumping in Krishnagiri and Salem: https://www.thenewsminute.com/article/tn-farmers-dump-tomato-harvests-lakes-after-prices-drop-77230
- SELCO Foundation cold storage landscape study: https://selcofoundation.org/?p=5217
- Solar cold storage and FPO pay-per-use (Mongabay India): https://india.mongabay.com/2025/11/solar-powered-cold-storage-empowers-smallholder-farmers/
- IIM-Indore study report (news): https://www.freepressjournal.in/amp/indore/cold-storage-alone-cant-stop-indias-food-loss-crisis-iim-i-study
- State-wise cold storage capacity table (RBI): https://website.rbi.org.in/web/rbi/-/publications/table-103-state-wise-cold-storage-capacity
- Mandi price dataset (data.gov.in): https://data.gov.in/resource/current-daily-price-various-commodities-various-markets-mandi
- Reported past IdeaRun grant: https://mgmits.ac.in/ec-news-events/participated-and-granted-2-982-in-the-ieee-smart-village-idearun-2024-national-finals

Prepared with AI assistance from public web sources; search results were summaries, and several source pages were not read in full. Check every figure against the original before you submit.